INCH-POUND

MIL-C-24643/40C
14 March 1994
SUPERSEDING
MIL-C-24643/40B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1SMU
(INCLUDING VARIATION LS1SMA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LS1SMU (unarmored) and LS1SMA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

	First	- Copper conductor, tin coated, ASTM B 286, size 22-7. Maximum diameter 0.033 inch.
	Second	- Separator may be used at manufacturer's option where required to provide free-stripping insulation.
*	Third	- Cross-linked polyethylene insulation. Nominal wall thickness 0.030 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.
	Fourth	- Braided shield of tin coated copper. Minimum coverage of 85 percent.
*	Fifth	- Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed. An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

AMSC N/A FSC 6145

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Construction (non-watertight) (continued)

* Sixth - The required number of singles (see table I) cabled around a center filler with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used as necessary to form a firm well rounded cable.

Seventh - Binder tape, applied helically with overlap.

Construction variant for type LS1SMU

* Eighth - Cross-linked polyolefin jacket. Minimum average thickness 0.063 inch.

Construction variant for type LS1SMA

* Eighth - Cross-linked polyolefin jacket. Minimum average thickness 0.063 inch.

* Ninth - Braided metal armor.

TABLE I. Details.

Military 1/ part no. M24643/40	Size 2/	Number of Conductors	Diameter over completed cndct (inch) (nominal)	Overall diameter (inches)			
				TYPE LS1SMU		TYPE LS1SMA	
				min	max	min	max
-01UN	5	5	0.131	0.465	.500	.515	.550

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.

2/ Type LS1SMU and LS1SMA.

EXAMINATION AND TESTS:	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	17.37
Voltage withstand - volts, root mean square, minimum	
Conductor to shield	5000
Shield to shield	500
Insulation resistance - Megohms/1000 feet minimum	
* Conductor to shield	500
* Component shield to shield	100
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Characteristic impedance - at 1MHz, ohms	50 ± 2

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EXAMINATION AND TESTS (continued):	Requirements
*Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Attenuation - at 400 MHz, decibel (db)/100 FT maximum	18
Cold bending, cable - (see table I for mandrel diameter)	No damage
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals - (aged)	
Insulation	
Air Oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air Oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
* Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - (types LS1SMU only) cycles, minimum	125
Armor (type LS1SMA only) - conformance to material, construction and coverage	No failure
* Group D:	
Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable)(125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
* Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1000 feet nominal.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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